Question	Answer	Marks	Guidance
5(a)(i)	$\omega = 2\pi / 1.7 = 3.7$ rad s^{-1}	A1	Working and answer needed. Unit not needed, but any unit given must be correct.
5(a)(ii)	(magnitude of) maximum displacement (of block) from equilibrium position	B1	**Allow** 'mean position' for 'equilibrium position'. **Do not allow** 'rest position' for 'equilibrium position'.
5(a)(iii)	$E = ½ m\omega^2 x_0^2$	C1	
	$13 \times 10^{-6} = ½ \times 27 \times 10^{-3} \times 3.7^2 \times x_0^2$	C1	No ECF from **5(a)(i)**.
	amplitude = 8.4×10^{-3} m	A1	Correct to at least 2 significant figures. AFC. (8.37)
5(b)	dome-shaped curve with correct curvature and a single peak with maximum E_K shown at 13 μJ	B1	**Allow** tolerance ±½ small square vertically.
	curve symmetrical about E_K-axis, with single peak at $x = 0$, end points at $E_K = 0$ and x-axis intercepts at the same positions in the positive and negative directions	B1	**Allow** tolerance ±1 small square horizontally. Vertically, end-points must get to within ½ small square of the x.axis. Does not need to have the correct curvature but must have a single peak and end points at $E_K = 0$.
	x-axis scale marked with a consistent scale that corresponds with the end points of the line being at ± values of x equal to the answer in **5(a)(iii)**	B1	At least one scale marking needed in each half (positive and negative) of the x-axis. All scale markings given must be consistent with the scale to within ½ small square. Scale does not need to conform to practical exam standards, it just needs to convey the correct intercept values to a ½ small square tolerance. Line does not need to be dome-shaped for this marking point.